Lesson Planning

Complete Lesson Plan Book

Class: FSc Pre-Medical · Class 11

Subject: Physics

Medium: English

Academic Year 2026-27

DM

This book is prepared as a teaching guide for teachers.

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Unit 1: Scalars and Vectors in Everyday Motion

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Learning Objectives

• Students will classify everyday quantities as scalar or vector with reasons.

• Students will explain why direction matters for vector quantities but not for scalar quantities.

• Students will represent a displacement with an arrow showing magnitude and direction.

Key Concepts

• A scalar quantity has magnitude only, for example distance, speed, mass and time.

• A vector quantity has both magnitude and direction, for example displacement, velocity, force and acceleration.

• Distance is the total path covered; displacement is the straight-line change of position from start to finish.

• A vector is drawn as an arrow: its length shows magnitude and its head shows direction.

Teaching Methodology (Step by Step)

1. Ask students to walk from the door to their desk and back, and discuss distance covered versus displacement.

2. Sort everyday quantities on the board into two columns: magnitude only, and magnitude plus direction.

3. Draw a displacement arrow for a student moving 3 metres east, then 4 metres north.

4. Compare speed and velocity using the example of a car going around a roundabout.

5. Work two short classification questions together, then let students answer three alone.

Classroom Activities

• Arrow walk: one student walks a path while the class draws the displacement arrow.

• Quantity cards: groups sort cards into scalar and vector piles and defend one tricky card.

• Everyday list: students list three scalars and three vectors from their own daily routine.

Assessment Questions

1. Is temperature a scalar or a vector? Give one reason.

2. A student walks 5 metres east and then 5 metres west. What is the displacement?

3. Draw an arrow for a force of 10 newtons acting towards the north.

Homework

• Write five quantities from daily life and label each scalar or vector with a reason.

• Draw the displacement arrow for a walk from your home gate to the nearest shop.

Time Allocation

40 minutes